

Wittgenstein Apartment

Sense-Level Behavioral Predicate Ontology

Abstract

This study introduces Wittgenstein Apartment, a behavioral predicate ontology developed to organize actions that humans can intentionally perform at the sense level for character creation and narrative planning. Wittgenstein Apartment is not an OWL/RDF ontology defined within formal ontology standards; rather, it organizes action predicates through its own classification and access layers. Its basic unit is not the lemma but the Action Sense, which represents a specific meaning of an action. The ontology is designed not to expose every theoretically possible action to every character, but to constrain a character's behavioral repertoire and thereby support consistency within a fictional setting. Action Senses are first placed within broad behavioral domains and Behavioral Predicate Families, then evaluated as Basic Human Actions or Character Specific Actions, and may ultimately receive character-level access states such as Allowed, Not Allowed, Conditional, or High-Cost Conditional. WordNet, VerbNet, and FrameNet provide lexical-semantic grounding and points of comparison, but Wittgenstein Apartment's Superclass and Family structures are not reproductions of the categories used by those resources, nor does the project claim to introduce a new linguistic classification on that basis. Its goal is to turn sense-level action information into a constrained behavioral repertoire that can be used in character creation and narrative planning, thereby supporting more consistent character behavior.

Keywords: behavioral predicate ontology; Action Sense; character repertoire; character creation; narrative planning; lexical semantics.

1. Introduction

Wittgenstein Apartment (WA) is a behavioral predicate ontology that organizes actions humans can intentionally perform at the sense level and is designed to make those actions usable in character creation and narrative planning. Here, the term ontology is not used in the sense of a formal OWL/RDF ontology. WA is a project-specific representation system that organizes action predicates under defined classes, relations, and access decisions. Its units are not nouns or adjectives but actions themselves; however, these actions are represented not at the lemma level, but at the Action Sense level, which distinguishes among different meanings of the same verb.

The central problem addressed by the project is the difference between an action being linguistically definable and that action actually being available within a particular character's behavioral repertoire. A character may understand the meanings of actions such as kill, operate, authorize, or steal, yet these actions need not be equally available to that character. Within a fictional setting, occupation, skills, social role, background, authority, and current circumstances may open some Action Senses while closing others. WA therefore does not aim to make the full range of human actions available to every character; it aims to constrain what a character can do in a systematic way.

To support this goal, WA organizes actions in a Superclass → Behavioral Predicate Family → Action Sense structure. Superclass represents a broad behavioral domain, Behavioral Predicate Family groups actions around a shared behavioral core, and Action Sense represents a specific action meaning. These concepts are not equivalent to synsets in WordNet, verb classes in VerbNet, or frames in FrameNet. WA can draw on lexical-semantic information from these resources and can be compared with them, but its own Superclass and Family structures are designed around the needs of character repertoires and narrative planning. The project does not present these structures as alternatives to those resources or as a new linguistic classification intended to replace them.

The presence of an Action Sense in the ontology does not mean that the action is available to every character. Before an Action Sense reaches the character level, it passes through several decision layers. The distinction between Basic Human Action and Character Specific Action indicates whether the action can be assumed in an initial repertoire; character-specific evidence such as skills, occupation, role, authority, and background can then narrow or expand access. At the final character level, an action may be marked Allowed, Not Allowed, Conditional, or High-Cost Conditional. In this way, a character in narrative generation acts not from the full set of actions represented in the ontology, but from a defined behavioral space specific to that character.

Wittgenstein Apartment should therefore be understood at two levels. First, it is a representational approach that proposes constraining the action space available to characters through the BHA/CSA distinction and character-specific access decisions. Second, it is a data resource that provides the action material required to implement that approach. By organizing actions that humans can intentionally perform at the sense level, the dataset provides a candidate behavioral space on which a character-modeling system can operate. The resource is therefore not itself a running simulation; however, its structure was designed so that its actions can later be used in character creation, narrative planning, or simulation systems.

The aim of this approach is to constrain character behavior through a predefined and traceable repertoire rather than leaving it entirely to unrestricted decisions made at generation time. From the perspective of narrative planning, WA is not a complete planner that determines which action should be selected; instead, it is intended to provide a prior representation layer that organizes which actions can become candidates for planning for a given character. The remainder of the paper first explains this conceptual positioning and the project's Wittgensteinian inspiration, then describes the ontology's Action Sense, Family, BHA/CSA, context, and character-access layers.

2. Conceptual Positioning and Related Work

2.1. Wittgensteinian Inspiration: From the Limits of Language to the Limits of a Character's Behavioral World

The name Wittgenstein Apartment is primarily inspired by proposition 5.6 of Wittgenstein's *Tractatus Logico-Philosophicus*, commonly rendered as 'the limits of my language mean the limits of my world.' The project does not use this statement as a philosophical method or criterion of validation (Wittgenstein, 1922, §5.6). Instead, WA reframes it as a narrower question for character modeling: does the linguistic availability of an action imply that the action is available within a particular character's behavioral world?

WA answers this question in the negative. The action *kill* may exist in a character's vocabulary; the character may understand its meaning and refer to it within the narrative. Yet the action may never become available within that character's own repertoire. The same distinction appears in everyday life: the actions people understand conceptually are not identical to the actions they can actually perform, are authorized to perform, or regard as part of their own behavioral range. WA treats this difference as an explicit access problem at the character level.

For this reason, the ontology's Action Sense inventory is not transferred directly to a character. Superclass and Behavioral Predicate Family organize the position of an action within the ontology, while BHA/CSA and character-specific evidence constrain whether that action enters a character's repertoire. At the final stage, Allowed, Not Allowed, Conditional, and High-Cost Conditional decisions define the behavioral space accessible to the character. The role of the Wittgenstein reference is limited to making one distinction visible: the set of actions that can be named linguistically is not the same as the set of actions behaviorally available to a particular character.

2.2. Relationship to Lexical-Semantic Resources

Wittgenstein Apartment is not intended to replace existing lexical-semantic resources such as WordNet, VerbNet, FrameNet, or PropBank. Each of these resources provides a different kind of information about actions. WA instead attempts to relate such information to the question of whether a given action is available to a particular character.

WordNet is useful for distinguishing different senses of a word. WA benefits from this distinction because different senses of the same verb may correspond to different behaviors. For WA, however, identifying the sense of a verb is not sufficient. The further question is whether that sense belongs in a particular character's behavioral repertoire (Miller, 1995).

VerbNet classifies verbs according to syntactic patterns and participant roles (Kipper et al., 2008). FrameNet represents actions and situations through broader semantic frames and the roles associated with those frames (Fillmore et al., 2003). PropBank annotates predicate-argument structures as they occur in texts (Palmer et al., 2005). These resources are primarily concerned with the meaning and linguistic realization of actions.

WA focuses on a different question. Once an Action Sense has been identified, its availability to a character is evaluated separately. An action may be clearly defined in language while still requiring a particular skill, occupation, social role, authority, or prior experience before it is appropriate for a given character. Likewise, even when a character has access to an action, current scene or world conditions may prevent that action from being performed at that moment.

The relationship between WA and WordNet, VerbNet, FrameNet, or PropBank is therefore complementary rather than competitive. These resources describe the meaning and linguistic structure of actions, while WA focuses on how a given action enters a character's behavioral space. The goal is not to create a new lexical-semantic system, but to connect existing sense information to a behavioral structure that can be used in character creation and narrative planning.

WordNet-derived information is therefore kept separate within WA. This separation prevents external lexical provenance from being conflated with WA's own character-level behavioral decisions.

2.3. Relationship to Commonsense and Planning Systems

WA is related not only to lexical-semantic resources, but also to commonsense resources that model possible causes and consequences of events, and to planning systems that construct sequences of actions. These families of resources, however, answer different questions.

Commonsense resources such as ATOMIC, Event2Mind, and GLUCOSE provide information about likely intentions, consequences, and reactions associated with events (Sap et al., 2019; Rashkin et al., 2018; Mostafazadeh et al., 2020). Narrative planning and approaches such as HTN planning instead address which actions should be selected to achieve goals, what conditions are required, and how complex tasks can be decomposed into steps (Riedl & Young, 2010; Georgievski & Aiello, 2015).

WA addresses an earlier stage: once a particular action sense has been identified, which characters should have that action in their repertoire, and is the action executable under the current conditions? WA therefore does not generate a plan or predict the outcome of an event. Its purpose is to organize the action space available to a character before planning or narrative generation begins.

This positioning is not a claim that the literature lacks a relevant area of research. WA proposes a more limited representational distinction: decisions about what an action means, whether a character can access it, and whether it can be executed under current conditions are kept separate while remaining linked to the same Action Sense.

Table 1. Focus of related resource families and their relationship to WA

Resource approach /	Primary focus	Main difference from WA
WordNet	Represents different word senses and relations among those senses.	WA connects this sense information to a character's behavioral repertoire.
VerbNet	Describes verb classes, participant roles, and syntactic-semantic patterns.	WA focuses on whether a particular Action Sense is available to a character.
FrameNet	Models events and situations through semantic frames and their roles.	WA does not build frames; it keeps character access and contextual requirements separate.
PropBank	Annotates predicate-argument structures as they occur in texts.	WA stores reusable action records that are not tied to a particular text instance.
ATOMIC / Event2Mind / GLUCOSE	Represent likely intentions, consequences, reactions, and commonsense relations associated with events.	WA focuses on whether the action enters a character's repertoire before reasoning about its consequences.
Narrative planning	Generates event and action sequences in relation to character goals.	WA constrains which actions a character can access before plan generation.
HTN / symbolic planning	Makes actions executable through preconditions, effects, and task decomposition.	WA does not provide planner rules; it organizes the action records to which such rules can be attached.
Wittgenstein Apartment	Organizes sense-level actions in terms of character access and executability under current conditions.	Its main aim is to create a consistent and constrained behavioral repertoire for a character.

As the table shows, WA's contribution is not to argue that existing resources are incomplete. Its aim is to connect action information that can be drawn from different resources with a separate layer that constrains a character's behavioral repertoire. This is particularly relevant to character creation and narrative planning, where the goal is not to give a character theoretical access to every human action, but to limit access to actions consistent with that character's properties and narrative circumstances.

3. Wittgenstein Apartment's Behavioral Structure

Wittgenstein Apartment does not store human actions as a flat list of verbs. Because the same verb can carry different meanings, actions are organized at three levels. Superclass identifies the broad behavioral domain to which an action belongs; Behavioral Predicate Family groups closely related action meanings; and Action Sense represents the specific action meaning evaluated by the system.

For example, walk by itself is not sufficiently precise for the system. Walking somewhere, walking someone to a destination, and walking an animal may all involve the same verb, yet they correspond to different behaviors. WA therefore treats the specific meaning of the verb, rather than the verb form alone, as the relevant unit.

The following table summarizes the distinction among these three levels.

Level	Plain-language description	Example function
Superclass	Identifies the broad behavioral domain to which an action belongs.	Separates broad areas such as movement, communication, care, or combat.
Behavioral Predicate Family	Groups closely related action meanings into a shared behavioral family.	Groups related forms of walking, carrying, or using under a common behavioral category.
Action Sense	Identifies the specific action meaning evaluated for a character.	Allows different senses of the same verb to be treated as distinct behaviors.

Superclass and Family are used to organize actions and make them easier to locate. Character-level decisions, however, are made at the Action Sense level. Two actions may share a verb form or belong to the same behavioral family while still having different implications for a character.

3.1. Superclass, Family, and Action Sense

Superclass indicates the broad behavioral domain of an action. For example, LOCOMOTION concerns the movement of an actor's own body, TRANSPORT concerns changing the location of another person or object, and MANIPULATION concerns direct physical action on an object or the environment. Behavioral Predicate Family groups more closely related action meanings within these broad domains. Family is not intended as a new linguistic classification; it is a practical organizational level used within WA to group similar behaviors.

A Family may contain multiple Action Senses, and the same lemma may belong to different Families when used in different senses. Character-level decisions are therefore made on the basis of Action Sense rather than Family or lemma. An Action Sense represents a particular meaning of a verb. For example, operate may mean operating a machine, running a company, or performing surgery. A character may have access to one of these senses while lacking access to the others.

Each Action Sense is assigned an identifier. The identifier allows the same action record to be tracked across different tables without ambiguity; it carries no special ontological claim. An Action Sense record also stores information such as the dictionary form of the action, its definition, its classification, task structure, contextual requirements, body-object-world conditions, and typical participants. The following table summarizes the main functions of these fields.

Main information stored in an Action Sense record

Field	Function
Lemma	Provides the dictionary form of the action.
Sense Variant / Definition	Specifies the particular meaning intended.
Superclass	Indicates the broad behavioral domain of the action.
Family	Indicates the behavioral family to which the action belongs.
Task Structure	Indicates whether the action is a basic operation or a task composed of multiple steps.
Scene Dependency	Indicates whether the action requires a particular social, institutional, or relational context.
Affordance information	Records the bodily, object-related, and environmental conditions required for the action.
Action Anatomy	Indicates the people or objects that typically participate in the action.

4. Constraining a Character's Action Space: BHA and CSA

A central goal of Wittgenstein Apartment in character modeling is to avoid making every human action represented in the ontology available to every character. The fact that an action can be performed by a human does not mean it should belong to the behavioral repertoire of a particular character. Walking, speaking, or lifting an object may reasonably be available to many characters by default, whereas performing surgery, piloting an aircraft, or authorizing an institutional procedure may require additional information about training, occupation, role, experience, or authority.

WA therefore initially distinguishes Action Senses as Basic Human Actions (BHA) or Character Specific Actions (CSA). BHA refers to actions that can be assumed to belong to a repertoire even when little character-specific information is available. CSA covers actions whose availability depends on some character-specific property. This distinction does not indicate whether an action is easy or difficult, common or rare; it only indicates whether the action can reasonably enter the initial repertoire without additional information about the character.

The BHA/CSA distinction is not the final decision for a specific character. Once a character has been defined, occupation, skills, social role, background, authority, and related properties are considered, and the relevant Action Sense may be marked Allowed, Not Allowed, Conditional, or High-Cost Conditional for that character. The same action can therefore have different access states for different characters. Presence in the ontology defines the action; the character model determines which of those actions actually enter that character's repertoire.

These four states make the degree of access explicit. Allowed indicates ordinary access, Not Allowed indicates that the character lacks access, Conditional indicates access only under specified conditions, and High-Cost Conditional indicates that the action is possible but requires substantial cost, resistance, or exceptional conditions for that character.

State	Meaning
Allowed	The action is normally available to the character.
Not Allowed	The character does not have access to the action.
Conditional	The action becomes available only when specified conditions are met.
High-Cost Conditional	The action is possible, but entails substantial cost, resistance, or exceptional conditions for the character.

Access to an action must also be distinguished from the ability to perform it at a particular moment. Surgery may belong to a surgeon's repertoire, yet it cannot be performed if the necessary tools, setting, or patient are absent. Conversely, the presence of an operating room and the required equipment does not grant a medically untrained character access to surgery. WA therefore treats character repertoire and scene/world conditions as separate dimensions.

The purpose of this distinction is not to constrain characters arbitrarily. It is to create a behavioral space consistent with the character's background, skills, role, and position within the narrative. Narrative planning or character-creation systems can then operate over a repertoire that has already been constrained and can be justified, rather than selecting from the entire space of human actions.

5. Action Conditions, Goals, and Application Mapping

The presence of an action in a character's repertoire does not mean that it can be performed under every circumstance. Wittgenstein Apartment therefore stores information about the structure of an action, the context it requires, body-object-world conditions, goal relations, and possible application mappings separately from character access. These fields do not create new actions; they provide additional information about the same Action Sense.

Task Structure indicates whether an action is treated as a basic operation, a multi-step task, or a record that should not be treated as a task in the ordinary sense. Scene Dependency indicates whether the action requires a particular social, institutional, relational, or ritual context. A physically simple action, for example, may only be meaningful within a particular occupation or institution. These fields do not turn WA into an HTN or planning system; they simply expose structural and contextual properties that may later be connected to such a system (Georgievski & Aiello, 2015).

Affordance Gates record the bodily, object-related, and environmental conditions required for an action to be performed. Action Anatomy records the participants typically involved in the action. Some actions require only an agent, while others also require a target, recipient, object, or destination. Together, these fields help distinguish a character's access to an action from whether that action is executable in the current situation.

Goal Ontology stores the purposes with which an Action Sense may be associated in a separate table. The meaning of an action is not identical to the goal for which it is used: the same action can serve different goals in different situations, and some actions need not be linked directly to a goal at all. Goal relations are therefore kept separate from the definition of the Action Sense and can be used when needed by narrative-planning systems.

Functional Projection presents selected Action Sense records under application-oriented headings such as SPEAK, COMBAT, INSPECT, MOVE, MAKE, USE, TAKE, and GIVE. These headings do not alter the meaning of an action; they indicate the functional route through which a game, simulation, or narrative system might reach the relevant record. Different speech actions, for example, may be grouped under SPEAK while still resolving to their specific Action Sense records.

Table 6. Main types of information attached to an action record

Field	Plain-language question	Function
Task Structure	Is the action basic or composed of multiple steps?	Describes the task structure of the action.
Scene Dependency	Does the action require a particular social or institutional context?	Records the contextual requirements of the action.
Affordance Gates	Are the required bodily, object-related, and world conditions present?	Constrains whether the action is executable at that moment.
Action Anatomy	Which participants are involved in the action?	Records the expected person and object roles in the action.
Goal Ontology	With which goals can the action be associated?	Keeps relations between actions and goals separate from action identity.
Functional Projection	Through which function can an application reach this action?	Links actions to application-facing headings such as SPEAK, MOVE, or USE.

The numerical distributions of these fields are reported separately in the dataset and version documentation. The focus here is not on their counts, but on the kind of information each field adds to an action record.

6. Potential Applications and Limitations

WA is not itself a working planner, narrative engine, or validation system. Its structure can nevertheless support three broad application areas: planning and domain authoring, character-centered narrative and simulation, and rule-based validation of actions proposed by generative models. The present study does not experimentally evaluate performance in these uses; its purpose here is to clarify the kinds of systems to which the resource could be connected.

In planning systems, constructing domain models that define action preconditions, effects, and task relations can require substantial human effort. Recent work has investigated automating parts of this process and generating planning domains from natural-language descriptions (Oswald et al., 2024). WA does not provide a complete PDDL or HTN domain, but its Action Sense, task-structure, contextual, and participant information can serve as a starting point for linking an application's own operators, preconditions, effects, or decomposition rules to specific action records.

For character-centered narrative and simulation, WA's potential role is to constrain the action space available to a character before planning begins. Narrative-planning research uses planning methods to relate character goals to actions and to generate sequences of events (Riedl & Young, 2010; Cardona-Rivera et al., 2024). The BHA/CSA distinction indicates which actions require additional character information, while character-level access decisions specify which actions remain open to a particular character. Scene and world conditions can then further constrain whether those actions are executable at a given moment.

A similar structure could be used to validate actions proposed by large language models or other generative systems. Recent neuro-symbolic planning research has shown that generative models can be combined with explicit rules and symbolic verification (Shao et al., 2025). In such a system, WA could provide a basis for mapping a proposed action to an Action Sense and then checking character access and contextual conditions. The mapping and verification mechanism would still need to be developed separately; it is not part of the present resource.

The resource also has clear limitations. The current release is not a complete or universal catalogue of human behavior; its coverage reflects the lexical resources used, project-specific definitions, and the review process applied during development. The BHA/CSA distinction is likewise not a biological classification of human capabilities, but a repertoire decision intended for character creation. Some legacy Family identifiers, aliases, and incomplete WordNet or Goal links remain in the resource. In addition, the released core resource does not contain fully populated character profiles; character background, skills, world state, and narrative-specific rules must be defined by the application.

Table 7. Main application areas and additional components required

Application area	Information WA can provide	Additional components required
Planning / domain authoring	Provides Action Sense records together with task and contextual information.	Preconditions, effects, planner structure, and a world model must be defined separately.
Character / narrative systems	Provides BHA/CSA distinctions, character-access decisions, and executability conditions.	Character profiles, dynamic world state, and narrative logic must be defined separately.
LLM / rule-based validation	Provides Action Sense records and explicit behavioral constraints.	Action mapping, validation rules, and runtime integration must be developed separately.

7. Conclusion

Wittgenstein Apartment is a behavioral predicate ontology that organizes actions humans can intentionally perform at the sense level rather than the lemma level. Actions are first positioned within a Superclass, Behavioral Predicate Family, and Action Sense structure, but their presence in the ontology does not make them automatically available to every character.

The central character-modeling idea is to separate the meaning of an action from a character's access to that action. BHA and CSA indicate whether an action requires character-specific information before it enters the initial repertoire. At the character level, Allowed, Not Allowed, Conditional, and High-Cost decisions further constrain that repertoire. Scene, object, bodily, and world conditions then separately determine whether an action available to the character can actually be performed at a particular moment.

WA is not intended to replace WordNet, VerbNet, FrameNet, commonsense resources, or planning systems. Its purpose is to connect sense-level action information to a constrained and traceable behavioral repertoire that can be used in character creation and narrative planning. The system's central question is therefore not simply 'can a human perform this action?' but rather 'can this particular character access this particular action, and can it be performed under the current conditions?'

The present study does not provide an experiment demonstrating the performance superiority of this approach. The main direction for future work is therefore to evaluate how WA can be used in practical character-creation, narrative-planning, game-AI, and validation systems.

References

- Cardona-Rivera, R. E., Jhala, A., Porteous, J., & Young, R. M. (2024). The Story So Far on Narrative Planning. *Proceedings of the International Conference on Automated Planning and Scheduling*, 34(1), 489–499. <https://doi.org/10.1609/icaps.v34i1.31509>
- Fillmore, C. J., Johnson, C. R., & Petruck, M. R. L. (2003). Background to FrameNet. *International Journal of Lexicography*, 16(3), 235–250. <https://doi.org/10.1093/ijl/16.3.235>
- Georgievski, I., & Aiello, M. (2015). HTN planning: Overview, comparison, and beyond. *Artificial Intelligence*, 222, 124–156. <https://doi.org/10.1016/j.artint.2015.02.002>
- Kipper, K., Korhonen, A., Ryant, N., & Palmer, M. (2008). A large-scale classification of English verbs. *Language Resources and Evaluation*, 42(1), 21–40. <https://doi.org/10.1007/s10579-007-9048-2>
- Miller, G. A. (1995). WordNet: A Lexical Database for English. *Communications of the ACM*, 38(11), 39–41. <https://doi.org/10.1145/219717.219748>
- Mostafazadeh, N., Kalyanpur, A., Moon, L., et al. (2020). GLUCOSE: Generalized and Contextualized Story Explanations. *Proceedings of EMNLP 2020*, 4569–4586.
- Oswald, J., Srinivas, K., Kokel, H., Lee, J., Katz, M., & Sohrabi, S. (2024). Large Language Models as Planning Domain Generators. *Proceedings of the International Conference on Automated Planning and Scheduling*, 34(1), 423–431. <https://doi.org/10.1609/icaps.v34i1.31502>
- Palmer, M., Gildea, D., & Kingsbury, P. (2005). The Proposition Bank: An Annotated Corpus of Semantic Roles. *Computational Linguistics*, 31(1), 71–106. <https://doi.org/10.1162/0891201053630264>
- Rashkin, H., Sap, M., Allaway, E., Smith, N. A., & Choi, Y. (2018). Event2Mind: Commonsense Inference on Events, Intents, and Reactions. *Proceedings of ACL 2018*, 463–473.
- Riedl, M. O., & Young, R. M. (2010). Narrative Planning: Balancing Plot and Character. *Journal of Artificial Intelligence Research*, 39, 217–268.
- Sap, M., Le Bras, R., Allaway, E., Bhagavatula, C., Lourie, N., Rashkin, H., Roof, B., Smith, N. A., & Choi, Y. (2019). ATOMIC: An Atlas of Machine Commonsense for If-Then Reasoning. *Proceedings of AAAI*, 33, 3027–3035.
- Shao, J.-J., You, H.-J., Cai, G., Dai, Q., Dong, Z., & Guo, L.-Z. (2025). Breaking the Self-Evaluation Barrier: Reinforced Neuro-Symbolic Planning with Large Language Models. *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence*, 6129–6137. <https://doi.org/10.24963/ijcai.2025/682>
- Wittgenstein, L. (1922). *Tractatus Logico-Philosophicus* (C. K. Ogden, Trans.). Kegan Paul, Trench, Trubner & Co.
- Wittgenstein Apartment Behavioral Predicate Resource. (Working version). WITTGEINSTEIN APARTMENT MASTER.xlsx; 03_10 SEPERATER GROUP.xlsx; Behavioral Predicate Model documentation, Sections 2–11.